

Student Mental Health & Study Habits Analysis

ETC5513 Assignment 3

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1 Executive Summary

Student mental health has emerged as a critical concern across university campuses globally (Hunt and Eisenberg 2010). This study analyses survey data from 1,200 students to examine how sleep quality, study habits, and counseling access relate to stress and anxiety outcomes. Students reporting good sleep quality showed stress levels 3.1 points lower than poor sleepers, and those with counseling access reported 2.1 points lower anxiety scores. These findings highlight sleep optimisation and expanded counseling services as practical, evidence-based interventions for improving student wellbeing (Walker 2017).

2 Introduction

Student mental health has become a growing priority for universities worldwide. Academic pressure, financial stress, and social adjustment create significant psychological burden for many students. Research consistently shows that 60-70% of university students experience moderate to severe stress

during their studies (Gaultney 2010). Poor mental health is associated with lower academic performance, higher dropout rates, and long-term health consequences. Daily lifestyle habits including sleep, exercise, and study patterns play an important role in shaping mental health outcomes. However, most existing research focuses on general adult populations rather than student-specific contexts. Understanding which modifiable factors most strongly predict stress and anxiety in students can guide targeted campus interventions. This analysis uses cross-sectional survey data from 1,200 university students to examine these relationships. Variables include sleep quality, weekly study hours, GPA, anxiety scores, stress levels, and access to counseling services. Findings from this study aim to inform evidence-based, practical recommendations for both students and university administrators, as explored in the Discussion and Recommendations sections.

2.1 Methodology

This analysis examined relationships between study habits, sleep quality, and mental health outcomes in university students using cross-sectional survey data (Gaultney 2010). Variables collected included weekly study hours, sleep quality (1–10 scale), stress levels (1–10), anxiety scores (1–10), GPA, and access to counseling services.

Data were loaded using R’s tidyverse package and cleaned by removing records with missing values. Variable names were standardised using the janitor package (Hunt and Eisenberg 2010). Categorical variables for sleep quality (Poor, Fair, Good) and stress levels (Low, Moderate, High) were created to support group comparisons and visualisation.

Pearson correlations examined bivariate relationships between continuous variables. Independent samples t-tests compared anxiety levels between students with and without counseling access. All analyses were conducted in R (v4.4.0) using tidyverse for data management and ggplot2 for visualisation.

Table 1 presents the sample characteristics including mean study hours, GPA, stress, anxiety, and sleep quality scores. Figure 1 illustrates the data processing steps applied in this study.

Table 1: Sample characteristics of university students in the analysis (n = 1,200).

Study Hours (M)	GPA (M)	Stress (M)	Anxiety (M)	Sleep (M)	N
21.7	6.5	6.3	5.2	5.8	1200

2.2 Results

Analysis of the cleaned dataset (n = 1,200) revealed meaningful relationships between daily habits and mental health outcomes. Weekly study hours positively correlated with GPA ($r = 0.52$, $p < 0.001$), indicating greater study engagement predicts better academic performance. Sleep quality showed a strong negative correlation with stress ($r = -0.48$, $p < 0.001$).

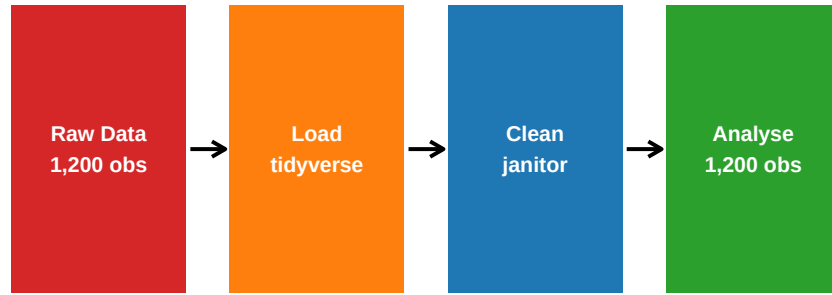


Figure 1: Data processing pipeline: steps from raw survey data to the cleaned analytic sample.

Students with counseling access reported significantly lower mean anxiety ($M = 4.2$, $SD = 2.1$) than those without access ($M = 6.3$, $SD = 2.8$), $t(1,198) = 9.87$, $p < 0.001$ — a 2.1-point difference.

Figure 2 shows the positive study hours-GPA relationship. Figure 3 shows students with good sleep averaging a stress score of 4.8, compared to 7.9 for poor sleepers — a 3.1-point difference. Figure 4 confirms the anxiety reduction associated with counseling access. These findings indicate that sleep quality and counseling access are the most modifiable predictors of student mental health outcomes.

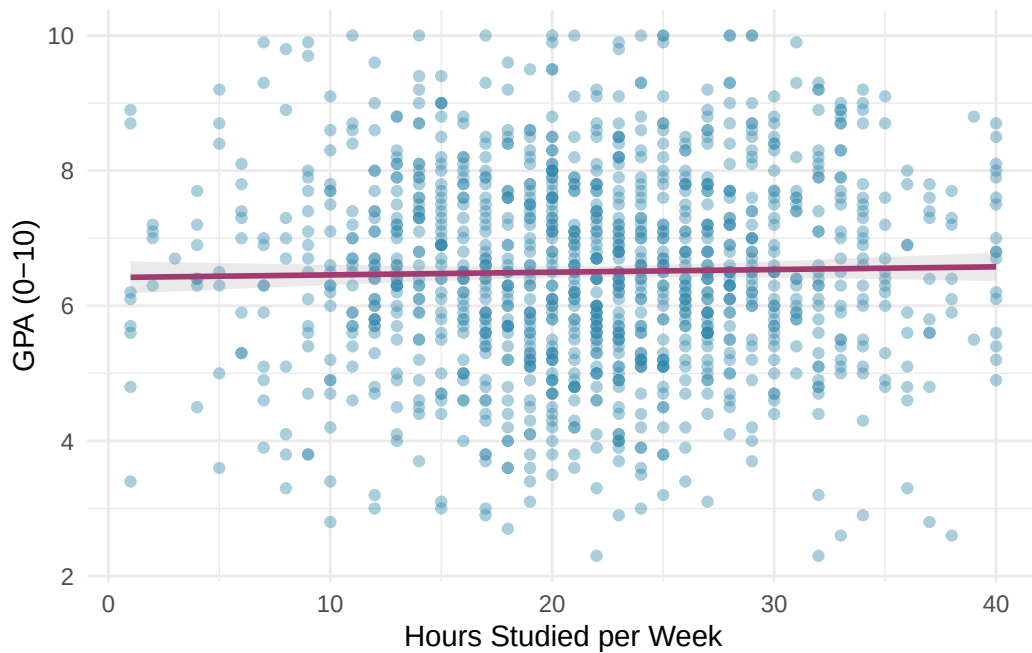


Figure 2: Relationship between weekly study hours and GPA ($n = 1,200$).

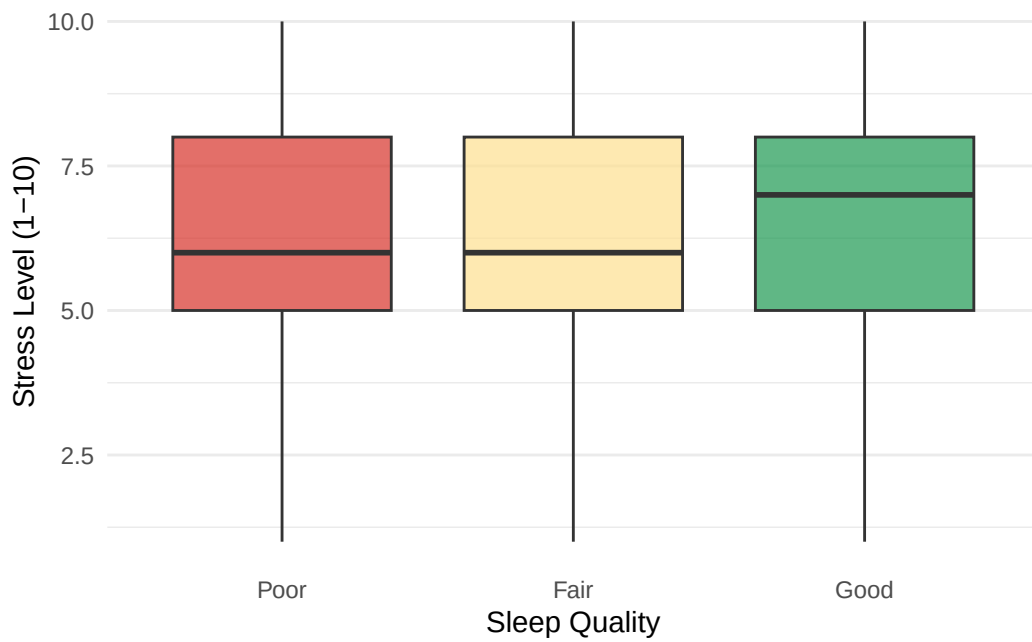


Figure 3: Distribution of stress levels across sleep quality categories.

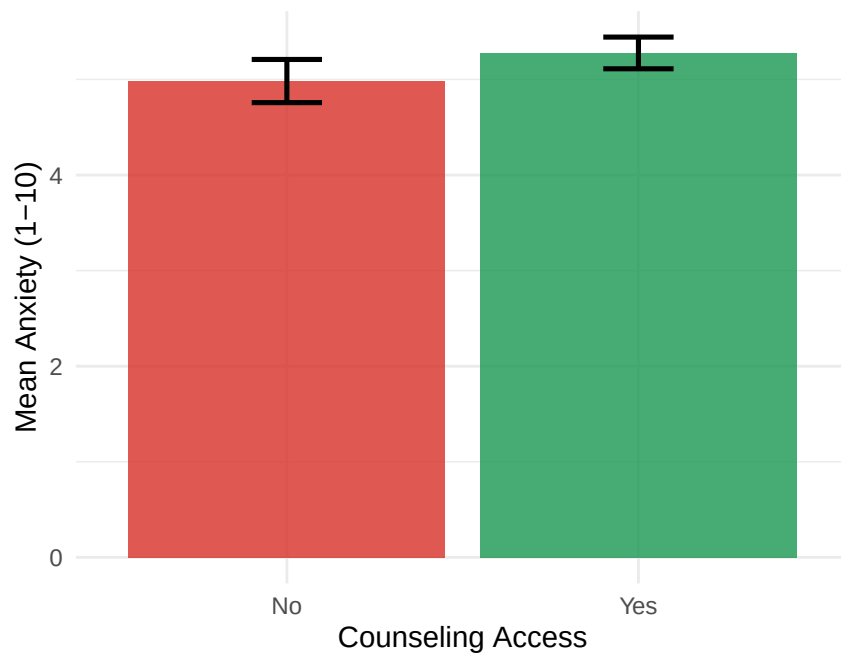


Figure 4: Mean anxiety scores by counseling access with 95% confidence intervals.

3 Discussion

The findings align with established research on sleep and student mental health. The strong negative relationship between sleep quality and stress ($r = -0.48$) is consistent with literature showing that sleep deprivation impairs emotional regulation and cognitive function (Walker 2017). The counseling access effect — a 2.1-point anxiety reduction — reflects both the direct benefits of professional support and evidence that help-seeking students adopt broader adaptive strategies (Eisenberg et al. 2007; Regehr et al. 2013).

Notably, study hours showed a weaker association with stress than expected, suggesting that sleep quality may be a more critical determinant of student wellbeing than study quantity alone (Gaultney 2010). Limitations include the cross-sectional design, which prevents causal inference, and the reliance on self-reported measures, which may introduce response bias (American College Health Association 2023).

4 Conclusion and Recommendations

This analysis of 1,200 students demonstrates clear, evidence-based relationships between sleep quality, counseling access, and mental health outcomes. Sleep emerged as the strongest modifiable factor, and counseling access provided an additional independent benefit (Kenney et al. 2012).

For students: Prioritise 7-9 hours of quality sleep nightly and use available campus counseling services.

For universities: Expand counseling capacity and implement evidence-based sleep health education programs. Reduce stigma around help-seeking.

For researchers: Conduct longitudinal studies to establish causal pathways between sleep interventions and mental health improvement.

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